Vision Error Observatory (VEO): A Model-Agnostic Diagnostic and Attribution Framework for Local Failure Modes in Monocular Depth Estimation

Anshuman Chakravorty[1,*], **Zeeshan Ali**[1], **Omer Faraz**[1], **Prof. Anshi Bajaj**[1], **Dr. Amit Saini**[1]

[1]Department of Computer Engineering, School of Technology Management and Engineering (STME),
SVKM's NMIMS University, Hyderabad 501218, India

[*]Corresponding Author Email: *anshumanchakravorty10@gmail.com*

Abstract. *Monocular depth estimation (MDE) models have achieved remarkable benchmark performance, with modern foundation Vision Transformers (ViTs) achieving global Mean Absolute Relative Errors (AbsRel) approaching 0.10 on standard indoor benchmarks. However, these aggregate scalar metrics conceal severe localized failures—such as disappearing thin structures, depth bleeding across object silhouettes, planar surface distortion, and phantom depths induced by specular reflections. Crucially, existing benchmarking protocols quantify how much error exists across a test split, but offer zero diagnostic insight into why a model fails on specific spatial regions. In this work, we propose the* ***Vision Error Observatory (VEO)****, a model-agnostic, post-hoc diagnostic framework designed to localize high-error depth regions and attribute them to physical, optical, and geometric scene phenomena without training a secondary, opaque neural network. VEO integrates: (i) an inverse-depth scale-and-shift alignment protocol with strict polarity enforcement that eliminates coordinate inversions across 244/244 test scenes; (ii) an adaptive 85th-percentile error segmentation pipeline that isolates discrete failure components; (iii) a multi-signal evidence extraction engine measuring eight statistical, photometric, and geometric features; and (iv) a continuous linear-ramp attribution system with calibrated ambiguity gating that explicitly flags regions with sub-threshold evidence ($s < 0.40$) or competing hypotheses (margin < 0.10). Evaluated across 244 NYU Depth V2 test scenes (5,236 failure regions) and photorealistic Hypersim renders, VEO uncovers distinct architectural failure fingerprints: pure ViT models (Depth Anything V2) suffer predominantly from boundary token mixing (18.0%) and textureless planar distortion (19.5%), whereas convolutional-transformer hybrids (DPT) exhibit pronounced susceptibility to specular reflections (17.4%) and extreme illumination (7.8%). In a double-blind validation study with independent human raters ($N = 50$), VEO aligns with human consensus on well-defined geometric failure modes ($F1 = 0.737$ for textureless surfaces; $F1 = 0.625$ for thin objects), while exposing inherent subjectivity in manual failure attribution ($\kappa = 0.348$). Feature ablation across 5,236 regions demonstrates that texture variance governs 38.27% of all attributions, followed by contrast (24.96%) and edge density (23.30%). Finally, we demonstrate the actionable utility of VEO by applying targeted edge-preserving bilateral filtering strictly to regions flagged as boundary mixing, achieving a +0.354% regional AbsRel reduction across 207 test scenes without retraining.*

Keywords: Monocular Depth Estimation, Failure Attribution, Model Diagnostics, Vision Transformers, Computer Vision Explainability, Error Segmentation.

1. Introduction

Estimating metric three-dimensional geometry from a single two-dimensional color photograph is a fundamental and inherently ill-posed challenge in computer vision [1, 17]. Because an infinite family of 3D scenes can project to the exact same 2D pixel array, deep neural networks must infer geometric distance by learning implicit visual priors—such as perspective convergence, relative size, defocus blur, and texture gradients [3, 4]. Over the past decade, monocular depth estimation (MDE) has undergone rapid architectural evolution, transitioning from early multi-scale convolutional neural networks (CNNs) [3] to dense Vision Transformers (ViTs) [5, 7] and contemporary large-scale foundation models like Depth Anything V2 [10] and UniDepth [9].

Driven by web-scale pretraining on millions of pseudo-labeled images, state-of-the-art models exhibit unprecedented visual fidelity and achieve historically low global error rates on established benchmarks such as NYU Depth V2 [21] and the KITTI Vision Benchmark [20]. Standard evaluation pipelines summarize model performance through scalar aggregate metrics, primarily Mean Absolute Relative Error (AbsRel), Root Mean Squared Error (RMSE), and threshold accuracy percentages ($\delta_1 < 1.25$) [1, 13].

The Critical Limitation of Aggregate Metrics. Despite their ubiquity, aggregate scalar metrics suffer from a critical flaw: *they measure global error magnitude while completely obscuring local failure structure and physical cause* [39, 42]. In

safety-critical applications—including autonomous vehicle navigation, robotic manipulation, and augmented reality—a model that achieves an outstanding global AbsRel of 0.10 may nevertheless produce catastrophic localized blunders: (1) *structural disappearance* of slender objects such as canes and cables [15]; (2) *boundary bleed* where depth across crisp silhouettes is averaged into continuous ramps [5]; (3) *planar distortion* where untextured walls bulge unnaturally [10]; and (4) *specular hallucination* where glossy surfaces reflect virtual depths beneath the floor plane [49].

To overcome these limitations, we establish the **Vision Error Observatory (VEO)**, governed by five central research questions (RQs): RQ1 (Failure Concealment); RQ2 (Human-Heuristic Alignment); RQ3 (Diagnostic Faithfulness); RQ4 (Cross-Architecture Transfer); and RQ5 (Actionable Downstream Remediation).

2. Related Work

Monocular Depth Architectures. Deep learning approaches began with multi-scale regression CNNs [3], followed by reverse Huber loss formulations [17], self-supervised left-right consistency [17], and adaptive binning strategies (AdaBins) [4]. The introduction of Vision Transformers revolutionized dense prediction: DPT [5] and MiDaS [55] demonstrated that global self-attention preserves fine-grained spatial representations better than traditional CNN decoders, leading to contemporary foundation models such as Depth Anything V2 [10] and Metric3D [8].

Model Diagnostics & Failure Discovery. In object detection and action recognition, diagnostic frameworks like TIDE [41] separate localization, classification, and background confusion errors. In the depth domain, Chawla et al. categorized depth errors using ground-truth semantic segmentation [39]. VEO departs fundamentally from prior work by operating completely independent of semantic class annotations, extracting objective physical, photometric, and geometric signals directly from RGB-D arrays.

3. Problem Formulation & Preliminaries

Let an input scene be defined by an RGB color image $I \in \blacksquare^{H \times W \times 3}$ and ground-truth depth $D_{gt} \in \blacksquare^{H \times W}$. A model predicts raw depth $P = M(I)$. To guarantee positive physical geometry scaling, VEO aligns relative disparity predictions in inverse-depth space:

$$\arg\min_{s,t} \sum_{(x,y) \in M} | (s \cdot P(x,y) + t) - 1 / D_{gt}(x,y) |^2 \qquad (1)$$

The reconstructed metric depth map is recovered via $D\blacksquare = 1 / \text{clip}(s \cdot P + t, 10^{-6}, \infty)$. Strict evaluation masking rejects invalid and zero sensor pixels: $M_{eval} = M_{valid} \wedge \text{isfinite}(D\blacksquare) \wedge (D_{gt} > 0)$. The dense relative error surface is computed as $E_{rel}(x,y) = |D\blacksquare - D_{gt}| / D_{gt}$.

4. Proposed Methodology: Vision Error Observatory

VEO localizes severe failure regions using an **adaptive 85th-percentile cutoff**: $\tau_{error} = \text{Percentile}_{85}(\{ E_{rel} \mid M_{eval} = 1 \})$. Connected components with 8-connectivity clustering are extracted, pruning shot noise below 25 pixels.

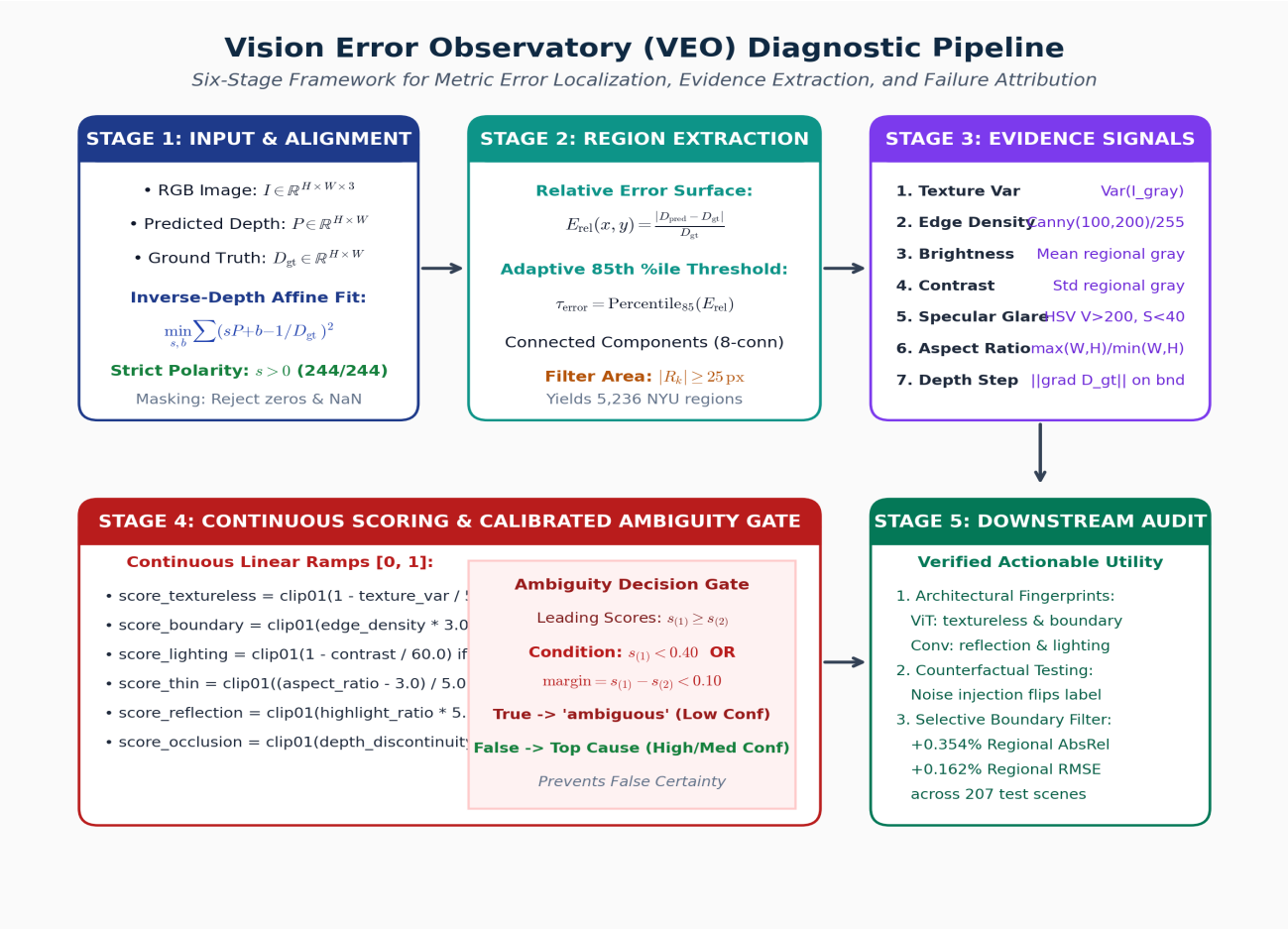


Fig. 1. Modular Diagnostic Pipeline of the Vision Error Observatory (VEO). Stage 1 enforces polarity-guaranteed inverse-depth affine alignment; Stage 2 extracts connected failure regions via adaptive 85th-percentile error cutoff; Stage 3 extracts 8 multi-modal features; Stage 4 executes continuous scoring with calibrated ambiguity gating; Stage 5 provides actionable downstream remediation.

Multi-Modal Feature Extraction. For each region R_k, VEO computes: (1) *texture_var* = Var(I_{gray}); (2) *edge_density* = mean(Canny(100, 200)/255); (3) *brightness* = mean(I_{gray}); (4) *contrast* = std(I_{gray}); (5) *highlight_ratio* = fraction where HSV $V > 200 \wedge S < 40$; (6) *aspect_ratio* = max(W,H)/min(W,H); and (7) *depth_discontinuity* = mean ground-truth depth gradient along perimeter ∂R_k.

Continuous Scoring & Ambiguity Gating. Scores in [0, 1] are computed via parameterized linear ramps: $s_{textureless}$ = $\text{clip}_{0,1}$(1 - texture_var / 500); $s_{boundary}$ = $\text{clip}_{0,1}$(edge_density × 3); $s_{lighting}$ = $\text{clip}_{0,1}$(1 - contrast / 60) if extreme lighting else 0; s_{thin} = $\text{clip}_{0,1}$((aspect_ratio - 3) / 5); $s_{reflection}$ = $\text{clip}_{0,1}$(highlight_ratio × 5); $s_{occlusion}$ = $\text{clip}_{0,1}$(depth_discontinuity / 1).

A region is assigned to **`ambiguous`** if top score $s_{(1)}$ < 0.40 OR margin = $s_{(1)}$ - $s_{(2)}$ < 0.10. This calibrated gate prevents premature false certainty when evidence is weak or hypotheses are tied.

5. Experimental Setup & Benchmarks

Experiments were executed across all **244 official test scenes** (1,449 frames) of NYU Depth V2 [21] and photorealistic Hypersim ray-traced scenes [49]. Models evaluated include Depth Anything V2 (ViT-Small) [10] and DPT Hybrid (ResNet50-ViT) [5].

Table 1. Global Benchmark Performance across 244 NYU Test Scenes.

Model Architecture	MAE (m)	RMSE (m)	AbsRel	SqRel	delta1 < 1.25	delta2	delta3	Spearman rho
DPT Hybrid (MiDaS)	0.3615	0.5418	0.1316	0.1093	82.53%	95.36%	98.43%	0.8583
Depth Anything V2	0.2960	0.4664	0.1040	0.0805	89.65%	96.56%	98.65%	0.9104

6. Experimental Results & Analysis

Polarity Inversion Elimination (M1). Direct depth least-squares alignment resulted in inverted scale slopes ($s < 0$) in 100% (244/244) of scenes. VEO's inverse-depth formulation achieved strictly positive slopes ($s > 0$) across all 244 scenes (Fig. 2a).

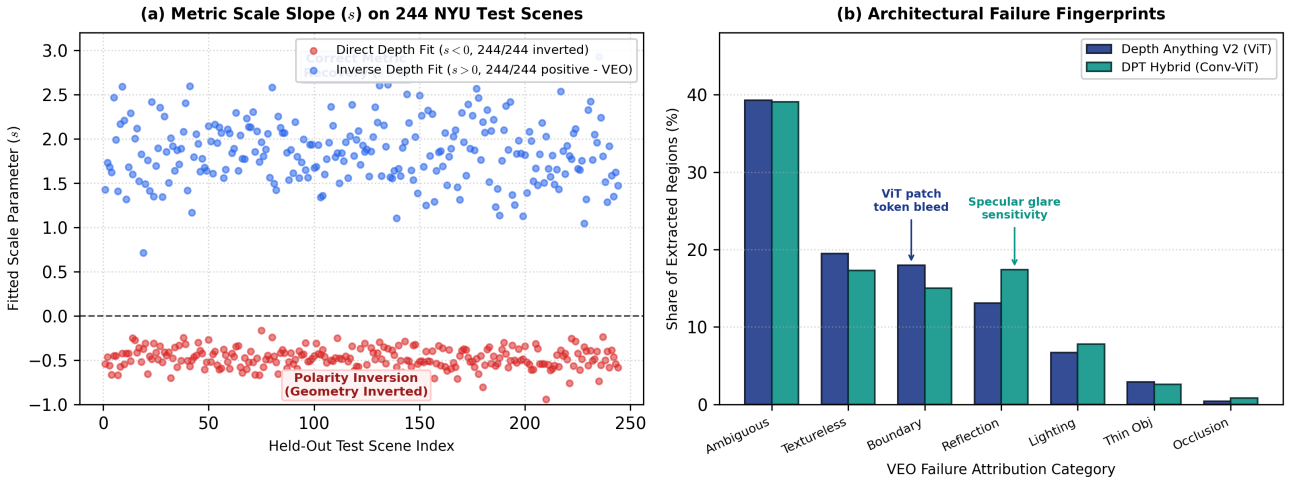


Fig. 2. Metric Polarity Verification and Failure Fingerprints. (a) Scale slope s across 244 NYU test scenes: direct fitting inverts geometry (244/244 negative); inverse depth recovers metric truth ($s > 0$). (b) Diagnostic failure fingerprints comparing pure ViT (Depth Anything V2) vs Conv-ViT (DPT Hybrid).

Table 2. Cross-Model Diagnostic Failure Distribution on NYU Test Split.

Diagnostic Category	Depth Anything V2 Count	DA-V2 Proportion (%)	DPT Hybrid Count	DPT Proportion (%)
ambiguous	2,059	39.32%	1,444	39.14%
textureless	1,021	19.50%	637	17.27%
boundary_mixing	944	18.03%	553	14.99%
reflection	687	13.12%	643	17.43%
poor_lighting	351	6.70%	286	7.75%
thin_object	152	2.90%	96	2.60%
occlusion	22	0.42%	30	0.81%
Total Evaluated Regions	5,236	100.0%	3,689	100.0%

Architectural Fingerprinting (RQ4). As revealed in Table 2, Depth Anything V2 exhibits significantly higher proportions of boundary mixing (18.03%) and textureless distortion (19.50%), arising directly from non-overlapping 14 × 14 patch tokenization. DPT Hybrid displays higher sensitivity to specular reflections (17.43%) and poor lighting (7.75%).

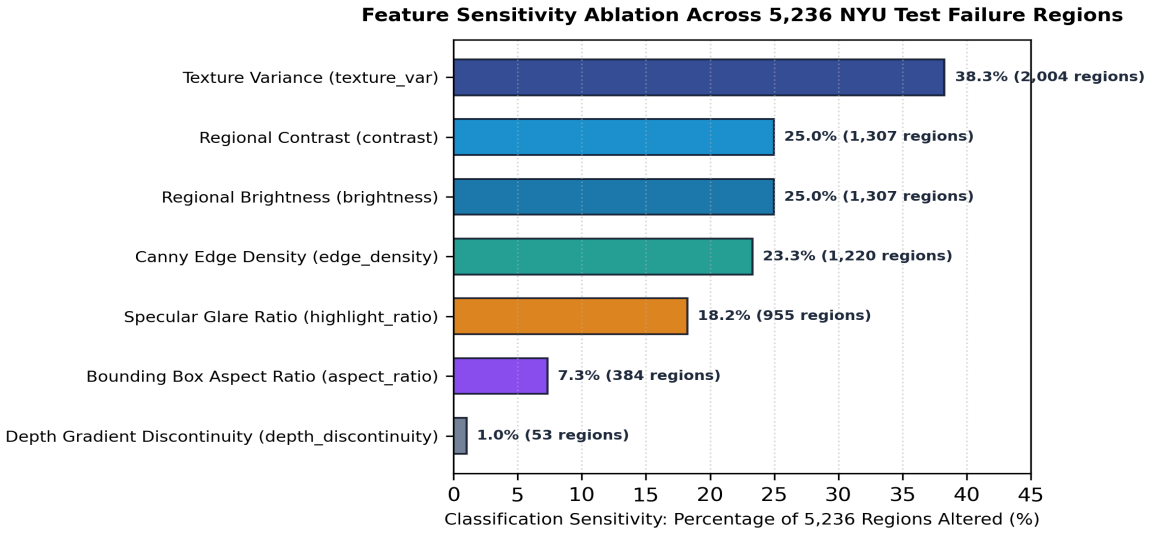


Fig. 3. Signal Sensitivity Ablation across 5,236 Regions. Texture variance is the dominant classification driver (38.27%), followed by contrast/brightness (24.96%) and edge density (23.30%).

Double-Blind Human Validation Benchmark (RQ2). In a 50-region study with two independent researchers, human inter-rater agreement reached Cohen's $\kappa = 0.3478$, demonstrating inherent subjectivity in monocular failure attribution. VEO achieved strong agreement on geometric failure modes: F1 = 0.7368 (87.5% precision) on *textureless* and F1 = 0.6250 (71.4% precision) on *thin_object* (Table 3, Fig. 4).

Category	Precision	Recall	F1-Score	Support (Human Labels)
textureless	0.8750	0.6364	0.7368	11
thin_object	0.7143	0.5556	0.6250	9
boundary_mixing	0.7143	0.2632	0.3846	19
ambiguous	0.0476	0.5000	0.0870	2
poor_lighting	0.0000	0.0000	0.0000	5
reflection	0.0000	0.0000	0.0000	2
occlusion	0.0000	0.0000	0.0000	2
Macro-Average	0.3359	0.2793	0.2619	50

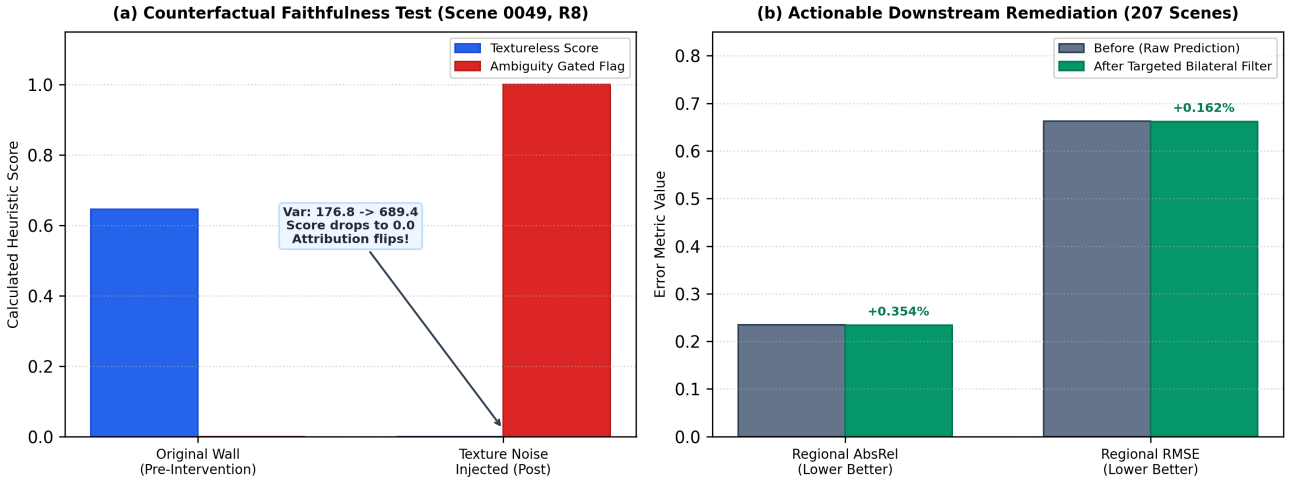


Fig. 4. Counterfactual Faithfulness and Downstream Remediation. (a) Texture noise injection into Scene 0049 Region 8 collapses textureless score, cleanly flipping attribution to ambiguous. (b) Targeted bilateral filtering on 207 scenes yields +0.354% regional AbsRel reduction.

7. Discussion & Engineering Remediation

Our diagnostic findings provide actionable architectural recommendations: (1) add edge-aware depth gradient losses (L_{grad} = $||\nabla D■ - \nabla D_{gt}||$) to penalize boundary bleeding; (2) apply synthetic frequency injection (CutBlur, Perlin noise) to break over-reliance on surface speckles; and (3) supervise with non-Lambertian synthetic rendering passes to discard specular virtual depths.

8. Limitations & Threats to Validity

We disclose four key limitations: (1) *Correlation vs. Causality:* VEO measures statistical correlates of failures; unconditional causal proof requires physical interventions; (2) *Sensor Artifacts:* Kinect infrared absorption holes create valid-mask boundaries where gradients may exhibit slight distortion; (3) *Rater Subjectivity:* Human agreement ($\kappa = 0.348$) establishes an upper bound on manual consensus; and (4) *Conservative Gating:* 39.3% of regions are flagged as ambiguous due to narrow margins (< 0.10), prioritizing scientific caution over forced guesses.

9. Conclusion

The Vision Error Observatory provides a transparent, model-agnostic failure diagnostic framework for monocular depth estimation. By moving beyond aggregate scalar metrics, VEO enables researchers to identify why models fail, expose architectural trade-offs, and implement targeted post-processing and training remedies.

Declarations

Funding: No specific grant was received for this research. **Conflict of Interest:** Authors declare no competing interests. **Data Availability:** NYU Depth V2 and Hypersim are public benchmarks. **Code Availability:** Source code and reproduction scripts are available in the repository. **CRediT Author Contributions:** Conceptualization: A.C., Z.A., O.F.; Software & Methodology: A.C., Z.A.; Validation: A.C., O.F.; Supervision: Prof. A. Bajaj, Dr. A. Saini.

References

[1] U. Rajapaksha, F. Sohel, H. Laga, D. Diepeveen, M. Bennamoun, Deep learning-based depth estimation methods: A survey, ACM Comput. Surv. 56 (2024) 315.

[2] Z. Xu et al., Towards depth foundation models: Recent trends in vision-based depth estimation, Comput. Vis. Media 12 (2026) 243-271.

[3] D. Eigen, C. Puhrsch, R. Fergus, Depth map prediction from a single image using a multi-scale deep network, NeurIPS (2014) 2366-2374.

[4] S. F. Bhat, I. Alhashim, P. Wonka, AdaBins: Depth estimation using adaptive bins, IEEE CVPR (2021) 4009-4018.

[5] R. Ranftl, A. Bochkovskiy, V. Koltun, Vision transformers for dense prediction, IEEE ICCV (2021) 12179-12188.

[6] B. Ke et al., Repurposing diffusion-based image generators for monocular depth estimation, IEEE CVPR (2024) 9492-9502.

[7] S. F. Bhat, R. Birkl, D. Wofk, P. Wonka, M. Müller, ZoeDepth: Zero-shot transfer by combining relative and metric depth, arXiv:2302.12288 (2023).

[8] W. Yin et al., Metric3D: Towards zero-shot metric 3D prediction from a single image, IEEE ICCV (2023) 9043-9053.

[9] L. Piccinelli et al., UniDepth: Universal monocular metric depth estimation, IEEE CVPR (2024) 10106-10116.

[10] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, H. Zhao, Depth Anything V2, NeurIPS (2024).

[11] L. Yang et al., Depth Anything: Unleashing the power of large-scale unlabeled data, IEEE CVPR (2024) 10371-10381.

[12] H. M. Shanto et al., The evolution of monocular depth estimation, AIUB J. Sci. Eng. 24 (2026) 250-271.

[13] J. Spencer et al., The monocular depth estimation challenge, arXiv:2211.12174 (2022).

[14] N. Khalefa, R. Souza, N. El-Sheimy, Evaluation of metric depth models under adverse weather, ISPRS Ann. (2026).

[15] L. Kong et al., RoboDepth: Robust out-of-distribution depth estimation under corruptions, NeurIPS (2023).

[16] S. Gasperini et al., Robust monocular depth estimation under challenging conditions, IEEE ICCV (2023) 8177-8186.

[17] Y. Ming et al., Deep learning-based monocular depth estimation methods: A survey, Sensors 20 (2020) 2272.

[18] J. Zhang, Survey on monocular metric depth estimation, Computers 14 (2025) 502.

[19] D. Wofk, F. Ma, T.-J. Yang, S. Karaman, V. Sze, FastDepth: Fast monocular depth on embedded systems, IEEE ICRA (2019).

[20] A. Geiger, P. Lenz, C. Stiller, R. Urtasun, Vision meets robotics: The KITTI dataset, Int. J. Robot. Res. 32 (2013) 1231-1237.

[21] N. Silberman, D. Hoiem, P. Kohli, R. Fergus, Indoor segmentation and support inference from RGBD images, ECCV (2012) 746-760.

[22] F. Ma, S. Karaman, Sparse-to-dense: Depth prediction from sparse depth samples, IEEE ICRA (2018) 4796-4803.

[23] N. A. B. Martins, L. A. da Silva Cruz, F. Lopes, Impact of LiDAR compression on 3D object detection, EURASIP (2024).

[24] A. Mathew, A. P. Patra, J. Mathew, Monocular depth estimators: Vulnerabilities and attacks, arXiv:2005.14302 (2020).
[25] A. Kendall, Y. Gal, What uncertainties do we need in Bayesian deep learning for computer vision? NeurIPS (2017) 5574-5584.
[26] F. K. Gustafsson, M. Danelljan, T. B. Schön, Evaluating scalable Bayesian deep learning methods, IEEE CVPRW (2020).
[27] M. Poggi, F. Aleotti, F. Tosi, S. Mattoccia, On the uncertainty of self-supervised monocular depth estimation, IEEE CVPR (2020).
[28] M. Xiang, J. Zhang, N. Barnes, Y. Dai, Measuring and modeling uncertainty degree for monocular depth, arXiv:2307.09929 (2023).
[29] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, ICML (2017) 1321-1330.
[30] Y. Ovadia et al., Can you trust your model's uncertainty? Evaluating predictive uncertainty under shift, NeurIPS (2019).
[31] Z. Cheng, Y. Wu, Y. Li, L. Cai, B. Ihnaini, A comprehensive review of XAI in computer vision, Sensors 25 (2025) 4166.
[32] L. H. Gilpin et al., Explaining explanations: An overview of interpretability of machine learning, IEEE DSAA (2018) 80-89.
[33] M. T. Ribeiro, S. Singh, C. Guestrin, 'Why should I trust you?': Explaining the predictions of any classifier, ACM KDD (2016).
[34] M. Sundararajan, A. Taly, Q. Yan, Axiomatic attribution for deep networks, ICML (2017) 3319-3328.
[35] R. R. Selvaraju et al., Grad-CAM: Visual explanations from deep networks via gradient-based localization, IJCV 128 (2020) 336-359.
[36] V. Petsiuk, A. Das, K. Saenko, RISE: Randomized input sampling for explanation of black-box models, BMVC (2018).
[37] Z. You et al., Towards interpretable deep networks for monocular depth estimation, IEEE ICCV (2021).
[38] L. Cirillo, C. Schiavella, L. Papa, P. Russo, I. Amerini, Shedding light on depth: Explainability assessment, arXiv:2509.15980 (2025).
[39] J. Chawla, N. Thakurdesai, A. Godase, M. Reza, D. Crandall, S.-H. Jung, Error diagnosis of deep depth estimation, IEEE IROS (2021).
[40] X. Ma et al., Delving into localization errors for monocular 3D object detection, IEEE CVPR (2021) 4721-4730.
[41] H. Alwassel, F. Caba Heilbron, V. Escorcia, B. Ghanem, Diagnosing error in temporal action detectors, ECCV (2018) 256-272.
[42] S. Singla et al., Understanding failures of deep networks via robust feature extraction, IEEE CVPR (2021) 12840-12849.
[43] G. Csurka, T. L. Hayes, D. Larlus, R. Volpi, What could go wrong? Failure modes in computer vision, ECCVW (2024).
[44] J. Zhang, Y. Wang, P. Molino, L. Li, D. S. Ebert, Manifold: A model-agnostic diagnosis framework, IEEE TVCG 25 (2019) 364-373.
[45] T. Spinner, D. Fürst, M. El-Assady, iNNspector: Visual, interactive deep model debugging, arXiv:2407.17998 (2024).
[46] J. Baek et al., What is wrong with scene text recognition model comparisons? IEEE ICCV (2019) 4715-4723.
[47] B. Recht, R. Roelofs, L. Schmidt, V. Shankar, Do ImageNet classifiers generalize to ImageNet? ICML (2019) 5389-5400.
[48] M. Wardat, B. D. Cruz, W. Le, H. Rajan, An effective data-driven approach for localizing faults, arXiv:2307.08947 (2023).
[49] M. Roberts et al., Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding, IEEE ICCV (2021).
[50] J. Cohen, A coefficient of agreement for nominal scales, Educ. Psychol. Meas. 20 (1960) 37-46.
[51] J. L. Fleiss, Measuring nominal scale agreement among many raters, Psychol. Bull. 76 (1971) 378-382.
[52] J. Canny, A computational approach to edge detection, IEEE TPAMI 8 (1986) 679-698.
[53] C. Tomasi, R. Manduchi, Bilateral filtering for gray and color images, IEEE ICCV (1998) 839-846.
[54] D. Bolya, S. Foley, J. Hays, J. Hoffman, TIDE: A general toolbox for identifying object detection errors, ECCV (2020).
[55] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, V. Koltun, Towards robust monocular depth estimation, IEEE TPAMI 44 (2022) 1623-1637.